Math 4441: Probability and Statistics

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Chapter 7
Parameter Estimation
(Part 2 – Interval Estimation)
Lecture 28

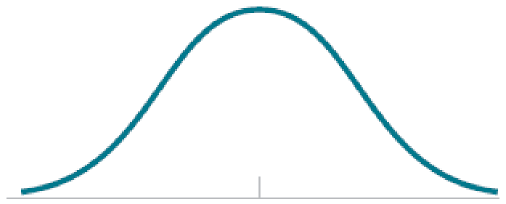
Math 4441: Probability and Statistics

References:

1. Sheldon Ross – Introduction to Probability and Statistics for Engineers and Scientists (6th Ed)
2. Douglas C Montgomery – Applied Statistics and Probability for Engineers (7th Ed)

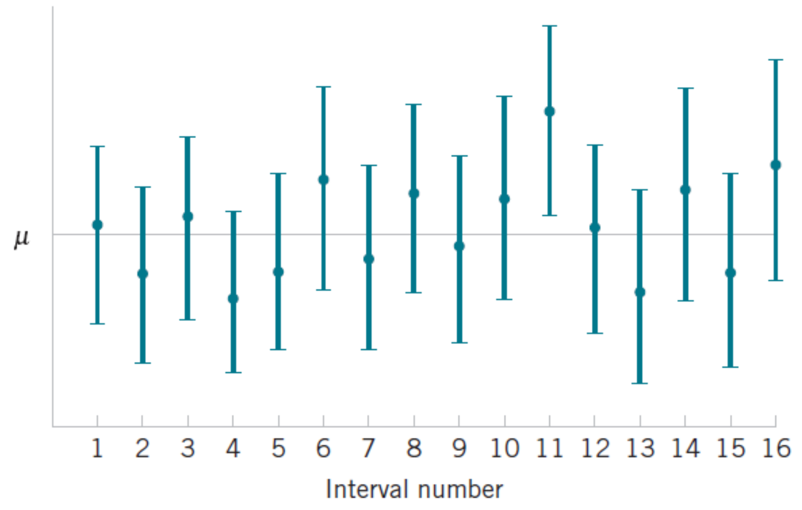
Confidence Interval

- A confidence interval is an estimation of an unknown parameter, where
  - The estimation is a Range of values
  - A confidence is associated with the estimation
- It describes the amount of uncertainty associated with a point estimate of a population parameter
- A confidence level indicates the percentage of time the confidence interval includes the true parameter value



Confidence Interval

- To find the confidence interval
  - Need to set the confidence level
    - 90%, 95%, 99%
- A 90% confidence level means
  - Estimated interval includes the true parameter value 90% of the time
- In general, we say $100(1-\alpha)\%$ CI
  - α is the significance level
  - Percentage of time the CI misses the parameter value


μ
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Interval number

Definition

- Let $X = [X_1\ X_2\ \dots X_n]^T$ be a random sample, with statistical parameter θ
- A $100(1-\alpha)\%$ CI for θ, with confidence level $(1-\alpha)$
  - is an interval $(\mathrm{L(X)},\ \mathrm{U(X)})$
  - With the property

$$P[L(X) < \theta < U(X)] = 1 - \alpha$$

- Estimate $(L(X), U(X))$ such that the interval includes the parameter value with a probability $1-\alpha$

Factors related to CI

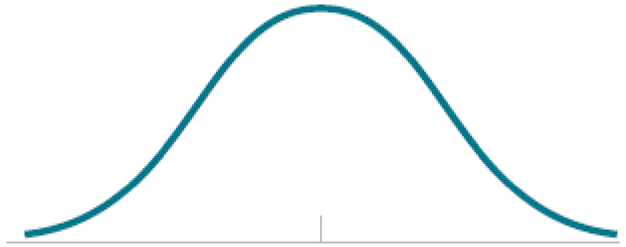
- The confidence level – value of $(1-\alpha)$
- The sample size – n
- The variability of the sample – sample or population variance σ^2
- The distribution of the estimator

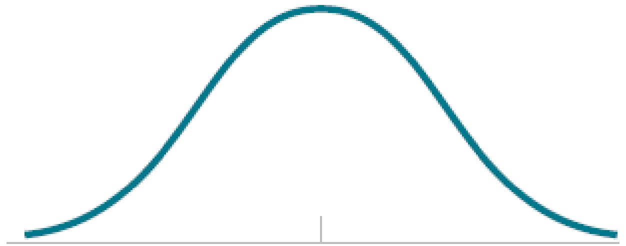
confidence interval
$$P(L(X) < \theta < U(X)) = \gamma$$

A higher confidence level would demand a wider confidence interval.

A larger sample would produce a narrower confidence interval.

Greater variability in the sample produces a wider confidence interval,





What exactly is the CI?

A confidence interval is the mean of your estimate

plus and minus

the variation in that estimate.

$$\boxed{\text{i.e., } CI = Mean \pm Variation}$$

This is the range of values you expect

your estimate to fall between

if you redo your test,

within a certain level of confidence.

Confidence, in statistics, is another way
to describe probability.

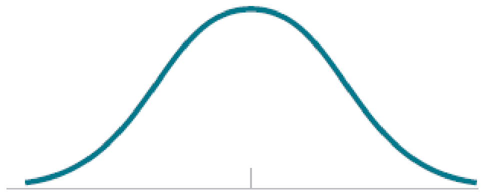
For example, if you construct a confidence interval
with a 95% confidence level,
you are confident that 95 out of 100 times
the estimate will fall between
the upper and lower values
specified by the confidence interval.

How to Calculate?

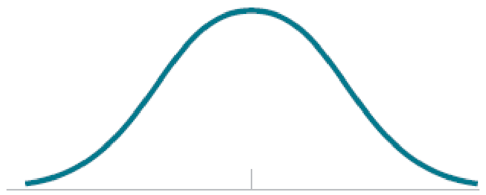
For calculating the confidence interval,

we need to know

- The point estimate (for which we need to construct the confidence interval)
- The critical values for the test statistic.
- The standard deviation of the sample.
- The sample size.



90% CI	$\alpha = 0.10, \quad \frac{\alpha}{2} = 0.05$	$z_{0.05} = 1.65$	
95% CI	$\alpha = 0.05, \quad \frac{\alpha}{2} = 0.025$	$z_{0.025} = 1.96$	
99% CI	$\alpha = 0.01, \quad \frac{\alpha}{2} = 0.005$	$z_{0.005} = 2.58$	
99% Upper CI	$\alpha = 0.01$	$z_{0.01} = 2.33$	



$$\frac{\overline{X}-\mu}{\sigma/\sqrt{n}} = \sqrt{n}\frac{(\overline{X}-\mu)}{\sigma}$$

has a standard normal distribution. Therefore,

$$P\left\{-1.96 < \sqrt{n}\frac{(\overline{X}-\mu)}{\sigma} < 1.96\right\} = .95$$

or, equivalently,

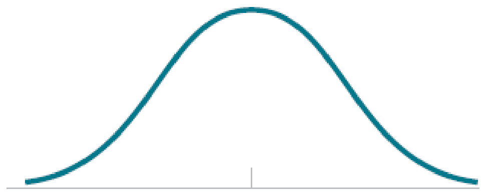
$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \overline{X}-\mu < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

Multiplying through by -1 yields the equivalent statement

$$P\left\{-1.96\frac{\sigma}{\sqrt{n}} < \mu-\overline{X} < 1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$

or, equivalently,

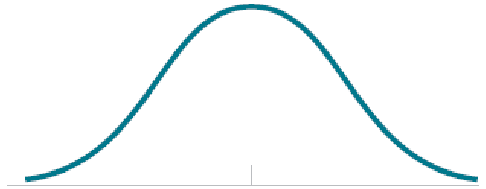
$$P\left\{\overline{X}-1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X}+1.96\frac{\sigma}{\sqrt{n}}\right\} = .95$$



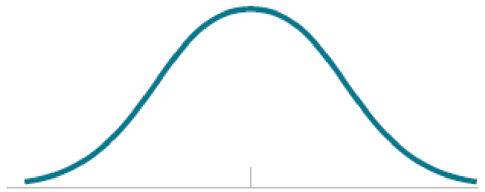
$$\overline{x} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}$$

$$\left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

is called a *95 percent confidence*



Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N, representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ.



$$\overline{x} = \frac{81}{9} = 9$$

a 95 percent confidence interval for μ is

$$\left(9 - 1.96\frac{\sigma}{3},\ 9 + 1.96\frac{\sigma}{3}\right) = (7.69, 10.31)$$

One-sided CI

$P\{Z < 1.645\} = .95$

As a result,

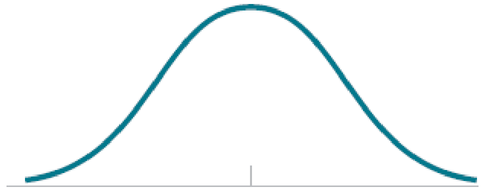
$$P\left\{\sqrt{n}\frac{(\overline{X}-\mu)}{\sigma} < 1.645\right\} = .95$$

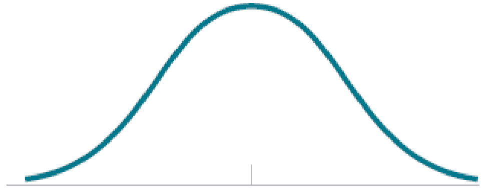
or

$$P\left\{\overline{X} - 1.645\frac{\sigma}{\sqrt{n}} < \mu\right\} = .95$$

$$\left(\overline{x} - 1.645\frac{\sigma}{\sqrt{n}}, \infty\right)$$

$$\left(-\infty, \overline{x} + 1.645\frac{\sigma}{\sqrt{n}}\right)$$





Example 7.3.a. Suppose that when a signal having value μ is transmitted from location A the value received at location B is normally distributed with mean μ and variance 4. That is, if μ is sent, then the value received is $\mu + N$ where N, representing noise, is normal with mean 0 and variance 4. To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, let us construct a 95 percent confidence interval for μ.

Example 7.3.b. Determine the upper and lower 95 percent confidence interval estimates of μ in Example 7.3.a.

$$1.645\frac{\sigma}{\sqrt{n}} = \frac{3.29}{3} = 1.097$$

the 95 percent upper confidence interval is

$$(9 - 1.097, \infty) = (7.903, \infty)$$

and the 95 percent lower confidence interval is

$$(-\infty, 9 + 1.097) = (-\infty, 10.097) \quad \blacksquare$$

Generalized $100(1-\alpha)\%$CI

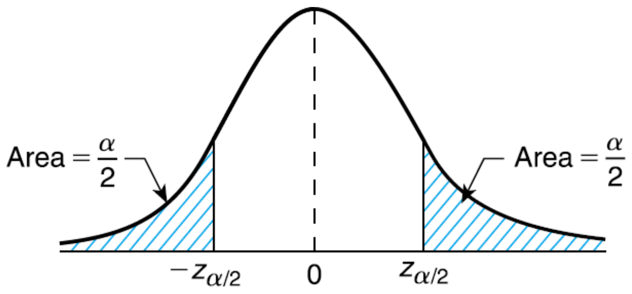
$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1-\alpha$$

As a result, we see that

$$P\left\{-z_{\alpha/2} < \sqrt{n}\frac{(\overline{X}-\mu)}{\sigma} < z_{\alpha/2}\right\} = 1-\alpha$$

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \overline{X}-\mu < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1-\alpha$$



$$P\{-z_{\alpha/2} < Z < z_{\alpha/2}\} = 1 - \alpha.$$

or

$$P\left\{-z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu - \overline{X} < z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

That is,

$$P\left\{\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\} = 1 - \alpha$$

$$\left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

$$\left(\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}},\ \infty\right)$$ 100(1 − α) percent one-sided upper

and

$$\left(-\infty,\ \bar{x} + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$$ 100(1 − α) percent

Example 7.3.c. Use the data of Example 7.3.a to obtain a 99 percent confidence interval estimate of μ, along with 99 percent one-sided upper and lower intervals.

Solution. Since $z_{.005} = 2.58$, and

$$2.58\frac{\alpha}{\sqrt{n}} = \frac{5.16}{3} = 1.72$$

it follows that a 99 percent confidence interval for μ is

$$9 \pm 1.72$$

Correction: $2.58\frac{\sigma}{\sqrt{n}} = \frac{5.16}{3} = 1.72$

since $z_{.01} = 2.33$

99 percent upper confidence interval is

$(9 - 2.33(2/3),\ \infty) = (7.447,\ \infty)$

99 percent lower confidence interval is

$(-\infty,\ 9 + 2.33(2/3)) = (-\infty,\ 10.553)$

compute an interval of length .1 with 99 percent confidence,

99 percent confidence interval for μ

$$\left(\overline{x} - 2.58\frac{\sigma}{\sqrt{n}},\quad \overline{x} + 2.58\frac{\sigma}{\sqrt{n}}\right)$$

$$(7.28,\ 10.72)$$

Hence, its length is

$$5.16\frac{\sigma}{\sqrt{n}}$$

Thus, to make the length to .1

$$5.16\frac{\sigma}{\sqrt{n}} = .1$$

or

$$n = (51.6\sigma)^2$$

Example 7.3.d. From past experience it is known that the weights of salmon grown at a commercial hatchery are normal with a mean that varies from season to season but with a standard deviation that remains fixed at 0.3 pound. If we want to be 95 percent certain that our estimate of the present season's mean weight of a salmon is correct to within ± 0.1 pound, how large a sample is needed?

A 95 percent confidence interval estimate

$$\mu \in \left(\overline{x} - 1.96\frac{\sigma}{\sqrt{n}},\ \overline{x} + 1.96\frac{\sigma}{\sqrt{n}}\right)$$

$\overline{x}$ is within $1.96(\sigma/\sqrt{n}) = .588/\sqrt{n}$

95 percent certain that $\overline{x}$ is within 0.1 of μ

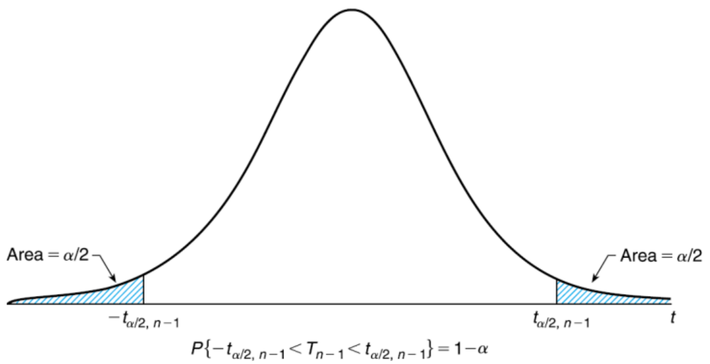
$$\frac{.588}{\sqrt{n}} \leq 0.1$$

That is, provided that

$$\sqrt{n} \geq 5.88$$

$$n \geq 34.57$$

CI for Unknown Variance



$\sqrt{n}(\overline{X} - \mu)/\sigma$ is a standard normal

$S^2 = \sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)$

$\sqrt{n}\,\dfrac{(\overline{X} - \mu)}{S}$ t-random variable with $n-1$ degrees of freedom

$$P\left\{-t_{\alpha/2,n-1} < \sqrt{n}\,\frac{(\overline{X}-\mu)}{S} < t_{\alpha/2,n-1}\right\} = 1-\alpha$$

or, equivalently,

$$P\{-t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}} < \bar{X}-\mu < t_{\alpha/2,n-1}\,\frac{S}{\sqrt{n}}\} = 1-\alpha$$

$$P\left\{\overline{X} - t_{\alpha/2,n-1}\frac{S}{\sqrt{n}} < \mu < \overline{X} + t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}\right\} = 1-\alpha$$

$$\mu \in \left(\overline{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}},\ \overline{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}\right)$$

TABLE V Percentage Points $t_{\alpha,\nu}$ of the t Distribution

ν \ α	.40	.25	.10	.05	.025	.01	.005	.0025	.001	.0005
1	.325	1.000	3.078	6.314	12.706	31.821	63.657	127.32	318.31	636.62
2	.289	.816	1.886	2.920	4.303	6.965	9.925	14.089	23.326	31.598
3	.277	.765	1.638	2.353	3.182	4.541	5.841	7.453	10.213	12.924
4	.271	.741	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	.267	.727	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	.265	.718	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	.263	.711	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	.262	.706	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	.261	.703	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	.260	.700	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	.260	.697	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	.259	.695	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	.259	.694	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	.258	.692	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	.258	.691	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	.258	.690	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	.257	.689	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	.257	.688	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	.257	.688	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	.257	.687	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850
21	.257	.686	1.323	1.721	2.080	2.518	2.831	3.135	3.527	3.819
22	.256	.686	1.321	1.717	2.074	2.508	2.819	3.119	3.505	3.792
23	.256	.685	1.319	1.714	2.069	2.500	2.807	3.104	3.485	3.767
24	.256	.685	1.318	1.711	2.064	2.492	2.797	3.091	3.467	3.745
25	.256	.684	1.316	1.708	2.060	2.485	2.787	3.078	3.450	3.725
26	.256	.684	1.315	1.706	2.056	2.479	2.779	3.067	3.435	3.707
27	.256	.684	1.314	1.703	2.052	2.473	2.771	3.057	3.421	3.690
28	.256	.683	1.313	1.701	2.048	2.467	2.763	3.047	3.408	3.674
29	.256	.683	1.311	1.699	2.045	2.462	2.756	3.038	3.396	3.659
30	.256	.683	1.310	1.697	2.042	2.457	2.750	3.030	3.385	3.646
40	.255	.681	1.303	1.684	2.021	2.423	2.704	2.971	3.307	3.551
60	.254	.679	1.296	1.671	2.000	2.390	2.660	2.915	3.232	3.460
120	.254	.677	1.289	1.658	1.980	2.358	2.617	2.860	3.160	3.373
∞	.253	.674	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

ν = degrees of freedom.

Example 7.3.e. Let us again consider Example 7.3.a but let us now suppose that when the value μ is transmitted at location A then the value received at location B is normal with mean μ and variance σ^2 but with σ^2 being unknown. If 9 successive values are, as in Example 7.3.a, 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, compute a 95 percent confidence interval for μ.

$\overline{x} = 9$

and

$$s^2 = \frac{\sum x_i^2 - 9(\overline{x})^2}{8} = 9.5 \qquad s = 3.082$$

as $t_{.025,8} = 2.306$, a 95 percent confidence interval for μ

$$\left[9 - 2.306\frac{(3.082)}{3},\ 9 + 2.306\frac{(3.082)}{3}\right] = (6.63, 11.37)$$

Upper and Lower CI

$$P\left\{\sqrt{n}\frac{(\overline{X}-\mu)}{S} < t_{\alpha,n-1}\right\} = 1 - c$$

or

$$P\left\{\overline{X}-\mu < \frac{S}{\sqrt{n}}t_{\alpha,\, n-1}\right\} = 1-\alpha$$

or

$$P\left\{\mu > \overline{X} - \frac{S}{\sqrt{n}}t_{\alpha,\, n-1}\right\} = 1-\alpha$$

$$\mu \in \left(\overline{x} - \frac{s}{\sqrt{n}}t_{\alpha,n-1},\ \infty\right)$$

$$\mu \in \left(-\infty,\ \overline{x} + \frac{s}{\sqrt{n}}t_{\alpha,\, n-1}\right)$$

Example 7.3.f. Determine a 95 percent confidence interval for the average resting pulse of the members of a health club if a random selection of 15 members of the club yielded the data 54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77. Also determine a 95 percent lower confidence interval for this mean.

$>d = c(54, 63, 58, 72, 49, 92, 70, 73, 69, 104, 48, 66, 80, 64, 77)$

$>l = \text{ mean}(d) - qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$

$>u = \text{ mean}(d) + qt(.975, 14) * \text{sqrt}(\text{var}(d)/15)$

$>c(l, u)$

[1] 60.8669366732691, 77.6663966600643

CI for Variance of Normal Population

If $X_1, \ldots, X_n$ is a sample from a normal distribution μ and σ^2

we can construct a confidence interval for σ^2

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$P\left\{\chi^2_{1-\alpha/2,n-1} \leq (n-1)\frac{S^2}{\sigma^2} \leq \chi^2_{\alpha/2,n-1}\right\} = 1-\alpha$$

$$P\left\{\frac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right\} = 1-\alpha$$

when $S^2 = s^2$, a $100(1-\alpha)$ percent confidence interval for σ^2

$$\left(\frac{(n-1)s^2}{\chi^2_{\alpha/2,n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2,n-1}}\right)$$

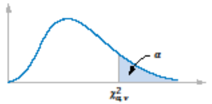


Table III Percentage Points $\chi^2_{\alpha,\nu}$ of the Chi-Squared Distribution

ν \ α	.995	.990	.975	.950	.900	.500	.100	.050	.025	.010	.005
1	.00+	.00+	.00+	.00+	.02	.45	2.71	3.84	5.02	6.63	7.88
2	.01	.02	.05	.10	.21	1.39	4.61	5.99	7.38	9.21	10.60
3	.07	.11	.22	.35	.58	2.37	6.25	7.81	9.35	11.34	12.84
4	.21	.30	.48	.71	1.06	3.36	7.78	9.49	11.14	13.28	14.86
5	.41	.55	.83	1.15	1.61	4.35	9.24	11.07	12.83	15.09	16.75
6	.68	.87	1.24	1.64	2.20	5.35	10.65	12.59	14.45	16.81	18.55
7	.99	1.24	1.69	2.17	2.83	6.35	12.02	14.07	16.01	18.48	20.28
8	1.34	1.65	2.18	2.73	3.49	7.34	13.36	15.51	17.53	20.09	21.96
9	1.73	2.09	2.70	3.33	4.17	8.34	14.68	16.92	19.02	21.67	23.59
10	2.16	2.56	3.25	3.94	4.87	9.34	15.99	18.31	20.48	23.21	25.19
11	2.60	3.05	3.82	4.57	5.58	10.34	17.28	19.68	21.92	24.72	26.76
12	3.07	3.57	4.40	5.23	6.30	11.34	18.55	21.03	23.34	26.22	28.30
13	3.57	4.11	5.01	5.89	7.04	12.34	19.81	22.36	24.74	27.69	29.82
14	4.07	4.66	5.63	6.57	7.79	13.34	21.06	23.68	26.12	29.14	31.32
15	4.60	5.23	6.27	7.26	8.55	14.34	22.31	25.00	27.49	30.58	32.80
16	5.14	5.81	6.91	7.96	9.31	15.34	23.54	26.30	28.85	32.00	34.27
17	5.70	6.41	7.56	8.67	10.09	16.34	24.77	27.59	30.19	33.41	35.72
18	6.26	7.01	8.23	9.39	10.87	17.34	25.99	28.87	31.53	34.81	37.16
19	6.84	7.63	8.91	10.12	11.65	18.34	27.20	30.14	32.85	36.19	38.58
20	7.43	8.26	9.59	10.85	12.44	19.34	28.41	31.41	34.17	37.57	40.00
21	8.03	8.90	10.28	11.59	13.24	20.34	29.62	32.67	35.48	38.93	41.40
22	8.64	9.54	10.98	12.34	14.04	21.34	30.81	33.92	36.78	40.29	42.80
23	9.26	10.20	11.69	13.09	14.85	22.34	32.01	35.17	38.08	41.64	44.18
24	9.89	10.86	12.40	13.85	15.66	23.34	33.20	36.42	39.36	42.98	45.56
25	10.52	11.52	13.12	14.61	16.47	24.34	34.28	37.65	40.65	44.31	46.93
26	11.16	12.20	13.84	15.38	17.29	25.34	35.56	38.89	41.92	45.64	48.29
27	11.81	12.88	14.57	16.15	18.11	26.34	36.74	40.11	43.19	46.96	49.65
28	12.46	13.57	15.31	16.93	18.94	27.34	37.92	41.34	44.46	48.28	50.99
29	13.12	14.26	16.05	17.71	19.77	28.34	39.09	42.56	45.72	49.59	52.34
30	13.79	14.95	16.79	18.49	20.60	29.34	40.26	43.77	46.98	50.89	53.67
40	20.71	22.16	24.43	26.51	29.05	39.34	51.81	55.76	59.34	63.69	66.77
50	27.99	29.71	32.36	34.76	37.69	49.33	63.17	67.50	71.42	76.15	79.49
60	35.53	37.48	40.48	43.19	46.46	59.33	74.40	79.08	83.30	88.38	91.95
70	43.28	45.44	48.76	51.74	55.33	69.33	85.53	90.53	95.02	100.42	104.22
80	51.17	53.54	57.15	60.39	64.28	79.33	96.58	101.88	106.63	112.33	116.32
90	59.20	61.75	65.65	69.13	73.29	89.33	107.57	113.14	118.14	124.12	128.30
100	67.33	70.06	74.22	77.93	82.36	99.33	118.50	124.34	129.56	135.81	140.17

ν = degrees of freedom.

Example 7.3.i. A standardized procedure is expected to produce washers with very small deviation in their thicknesses. Suppose that 10 such washers were chosen and measured. If the thicknesses of these washers were, in inches,

.123	.133
.124	.125
.126	.128
.120	.124
.130	.126

90 percent confidence interval for the standard deviation

$>d = c(.123, .124, .126, .120, .130, .133, .125, .128, .124, .126)$

$>l = 9 * \text{ var}(d)/\text{ qchisq}(.95, 9)$

$>u = 9 * \text{ var}(d)/\text{ qchisq}(.05, 9)$

$>c(l, u)$

$[1]7.26403231164731e-06 \quad 3.69611516361794e-05$

It follows that, with confidence .90,

$\sigma^2 \in (7.2640 \times 10^{-6}, \;\; 36.9612 \times 10^{-6})$

Taking square roots yields, with confidence .90,

$\sigma \in (.00269, \;\; .00608)$ ■

CI for the Differences in Means of 2 Normal Populations

$X_1, X_2, \ldots, X_n$ be a sample of size n from a normal population
mean μ_1 and variance σ_1^2

$Y_1, \ldots, Y_m$ be a sample of size m mean μ_2 and variance σ_2^2

interested in estimating $\mu_1 - \mu_2$.

$\overline{X} = \sum_{i=1}^{n} X_i / n$ and $\overline{Y} = \sum_{i=1}^{m} Y_i / m$

$\overline{X} - \overline{Y}$ maximum likelihood estimator of $\mu_1 - \mu_2$

$$\overline{X} \sim \mathcal{N}(\mu_1, \sigma_1^2 / n)$$
$$\overline{Y} \sim \mathcal{N}(\mu_2, \sigma_2^2 / m)$$

sum of independent normal random variables

$$\overline{X} - \overline{Y} \sim \mathcal{N}\left(\mu_1 - \mu_2, \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}\right)$$

assuming σ_1^2 and σ_2^2 are known

$$\frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} \sim \mathcal{N}(0, 1)$$

$$P\left\{ -z_{\alpha/2} < \frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{m}}} < z_{\alpha/2} \right\} = 1 - \alpha$$

$$P\left\{\overline{X}-\overline{Y}-z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n}+\frac{\sigma_2^2}{m}}<\mu_1-\mu_2<\overline{X}-\overline{Y}+z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n}+\frac{\sigma_2^2}{m}}\right\}=1-\alpha$$

two-sided confidence interval estimate for $\mu_1-\mu_2$ is

$$\mu_1-\mu_2\in\left(\overline{x}-\overline{y}-z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n}+\frac{\sigma_2^2}{m}},\ \overline{x}-\overline{y}+z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n}+\frac{\sigma_2^2}{m}}\right)$$

$100(1-\alpha)$ percent one-sided

$$\mu_1-\mu_2\in\left(-\infty,\ \overline{x}-\overline{y}+z_{\alpha}\sqrt{\sigma_1^2/n+\sigma_2^2/m}\right)$$

Example 7.4.a. Two different types of electrical cable insulation have recently been tested to determine the voltage level at which failures tend to occur. When specimens were subjected to an increasing voltage stress in a laboratory experiment, failures for the two types of cable insulation occurred at the following voltages:

Type A		*Type B*	
36	54	52	60
44	52	64	44
41	37	38	48
53	51	68	46
38	44	66	70
36	35	52	62
34	44		

$>l = \text{mean}(x) - \text{mean}(y) - \text{qnorm}(.975) * \text{sqrt}(40/14 + 100/12)$

$>u = \text{mean}(x) - \text{mean}(y) + \text{qnorm}(.975) * \text{sqrt}(40/14 + 100/12)$

$>c(l, u)$

[1]6.491114   19.604124

population variances σ_1^2 and σ_2^2 are unknown

$$S_1^2 = \sum_{i=1}^{n} \frac{(X_i - \overline{X})^2}{n-1}$$

$$S_2^2 = \sum_{i=1}^{m} \frac{(Y_i - \overline{Y})^2}{m-1}$$

the estimator of σ^2 is the *pooled estimator*

$$S_p^2 = \frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}$$

and the confidence interval is then based on the statistic

$$\frac{\overline{X} - \overline{Y} - (\mu_1 - \mu_2)}{\sqrt{S_p^2}\sqrt{1/n + 1/m}}$$

Assumption	Confidence Interval
σ_1, σ_2 known	$\overline{X} - \overline{Y} \pm z_{\alpha/2}\sqrt{\sigma_1^2/n + \sigma_2^2/m}$
σ_1, σ_2 unknown but equal	$\overline{X} - \overline{Y} \pm t_{\alpha/2,n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}$

Assumption	Lower Confidence Interval
σ_1, σ_2 known	$(-\infty, \overline{X} - \overline{Y} + z_{\alpha}\sqrt{\sigma_1^2/n + \sigma_2^2/m})$
σ_1, σ_2 unknown but equal	$\left(-\infty, \overline{X} - \overline{Y} + t_{\alpha,n+m-2}\sqrt{\left(\frac{1}{n} + \frac{1}{m}\right)\frac{(n-1)S_1^2 + (m-1)S_2^2}{n+m-2}}\right)$

Table 7.1 $100(1-\alpha)$ Percent Confidence Intervals

$$X_1, \ldots, X_n \sim \mathcal{N}(\mu, \sigma^2) \quad \overline{X} = \sum_{i=1}^{n} X_i/n, \qquad S = \sqrt{\sum_{i=1}^{n} (X_i - \overline{X})^2/(n-1)}$$

Assumption	Parameter	Confidence Interval	Lower Interval	Upper Interval
σ^2 known	μ	$\overline{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$	$\left(-\infty,\ \overline{X} + z_\alpha \frac{\sigma}{\sqrt{n}}\right)$	$\left(\overline{X} + z_\alpha \frac{\sigma}{\sqrt{n}},\ \infty\right)$
σ^2 unknown	μ	$\overline{X} \pm t_{\alpha/2,\, n-1} \frac{S}{\sqrt{n}}$	$\left(-\infty,\ \overline{X} + t_{\alpha,\, n-1} \frac{S}{\sqrt{n}}\right)$	$\left(\overline{X} - t_{\alpha,\, n-1} \frac{S}{\sqrt{n}},\ \infty\right)$
μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,\, n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\, n-1}}\right)$	$\left(0,\ \frac{(n-1)S^2}{\chi^2_{1-\alpha,\, n-1}}\right)$	$\left(\frac{(n-1)S^2}{\chi^2_{\alpha,\, n-1}},\ \infty\right)$